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Worksheet 2

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Semester: 3rd Semester **Date of Performance: 10-08-26**
Subject Name: Cloud Virtualization Lab **Subject Code: 25CAH-733**

Aim

To install, configure, and verify QEMU (Quick Emulator) on Ubuntu Linux and understand the basic concepts of virtualization by creating and managing virtual machine environments.

Objectives

- To understand the basic concepts of virtualization and virtual machine environments.
- To install and verify QEMU (Quick Emulator) on Ubuntu Linux.
- To create and manage files and directories using Linux commands.
- To gain practical experience in using QEMU for virtualization and system administration tasks.
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Software Used

- **Ubuntu Linux 24.04 LTS –** Host Operating System
- **QEMU (Quick Emulator) –** Virtualization and Emulation Software
- **QEMU Utilities (qemu-utils)** – Tools for creating and managing virtual disk images
- **Terminal (Command Line Interface) –** Used for executing Linux and QEMU commands

Procedure / Steps

- **Step 1:** Open the Ubuntu Terminal.
- **Step 2:** Update the package repository.
  sudo apt update
- **Step 3:** Install QEMU and QEMU Utilities.
  sudo apt install qemu-system-x86 qemu-utils -y
- **Step 4:** Verify the QEMU installation.
  qemu-system-x86_64 --version
- **Step 5:** Create a new directory for the practical.
  mkdir QEMU_Practical
- **Step 6:** Move to the created directory.

cd QEMU_Practical

- **Step 7:** Create a text file using UID and Name.
  Type UID_Name.txt
- **Step 8:** Add content to the file.
  echo "QEMU Practical Exercise" > 25mcc20044_mandeep.txt
- **Step 9:** Display the content of the file.
  cat 25mcc20044_mandeep.txt

Result:

```
C:\Users\akant>winget list QEMU
Name Id                                  Version Source
-----------------------------------------------------
QEMU SoftwareFreedomConservancy.QEMU 11.0.50 winget

C:\Users\akant>where /R "C:\Program Files" qemu-system-x86_64.exe
C:\Program Files\qemu\qemu-system-x86_64.exe
```

Fig 1 : QEMU version

```
C:\Users\akant>mkdir QEMU_Practical

C:\Users\akant>cd QEMU_Practical

C:\Users\akant\QEMU_Practical>touch 25mcc20044_Mandeep.txt
'touch' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\akant\QEMU_Practical>type nul > 25mcc20044_Mandeep.txt

C:\Users\akant\QEMU_Practical>dir
 Volume in drive C has no label.
 Volume Serial Number is 2698-9C50

 Directory of C:\Users\akant\QEMU_Practical

11-08-2026  15:10    <DIR>          .
11-08-2026  15:07    <DIR>          ..
11-08-2026  15:10                 0 25mcc20044_Mandeep.txt
               1 File(s)              0 bytes
               2 Dir(s)   3,216,031,744 bytes free
```

Fig 2: Folder Creation

```
C:\Users\akant\QEMU_Practical>echo QEMU Practical Exercise > 25mcc20044_Mandeep.txt

C:\Users\akant\QEMU_Practical>type 25mcc20044_Mandeep.txt
QEMU Practical Exercise
```

Fig 3: Display the content

Learning Outcomes:

- Understood the installation and verification process of QEMU virtualization software.
- Performed directory and file creation operations using command-line commands.
- Added and displayed file contents successfully.
- Developed hands-on experience in virtualization environment setup and basic system administration tasks.

Evaluation Grid :

Sr. No.	Parameters	Marks Obtained	Maximum Marks
1.	Worksheet Completion		08 marks
2.	Conduct		12 marks
3.	Viva Voice		10 marks
	Total		30 marks